

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the structure of a lysosome.

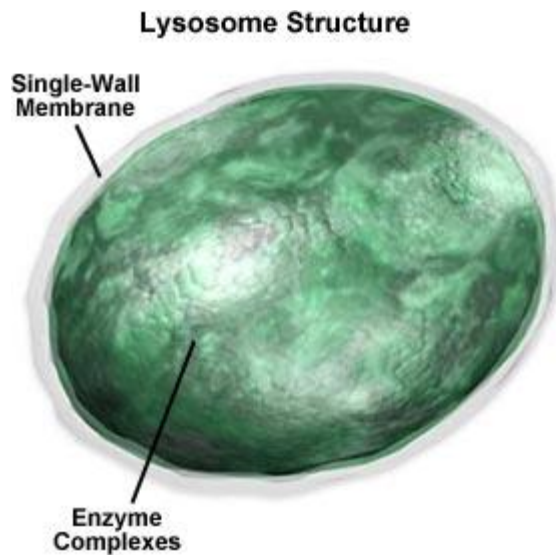


Fig. 1.1

- (a) State the main functions of the lysosome.

[2]

Tay-Sachs disease is caused by insufficient activity of the enzyme hexosaminidase A. Hexosaminidase A is a vital hydrolytic enzyme, found in the lysosomes, that breaks down glycolipids. When hexosaminidase A is no longer functioning properly, the lipids accumulate in the brain and interfere with normal biological processes.

Hexosaminidase A specifically breaks down fatty acid derivatives called gangliosides; these are made and biodegraded rapidly in early life as the brain develops. Patients with Tay-Sachs disease and carriers of Tay-Sachs can be identified by a simple blood test that measures hexosaminidase A activity.

- (b) (i) With reference to Fig. 1.1, outline how the enzyme hexosaminidase A in the lysosome is formed.

[4]

- (ii) Suggest **two** reasons for insufficient activity of the enzyme hexosaminidase A.

[2]

- (c) Explain how Tay-Sachs disease can affect nervous impulse transmission.

[2]

[Total: 10]

2 Fig. 2.1 shows the formation of collagen fiber.

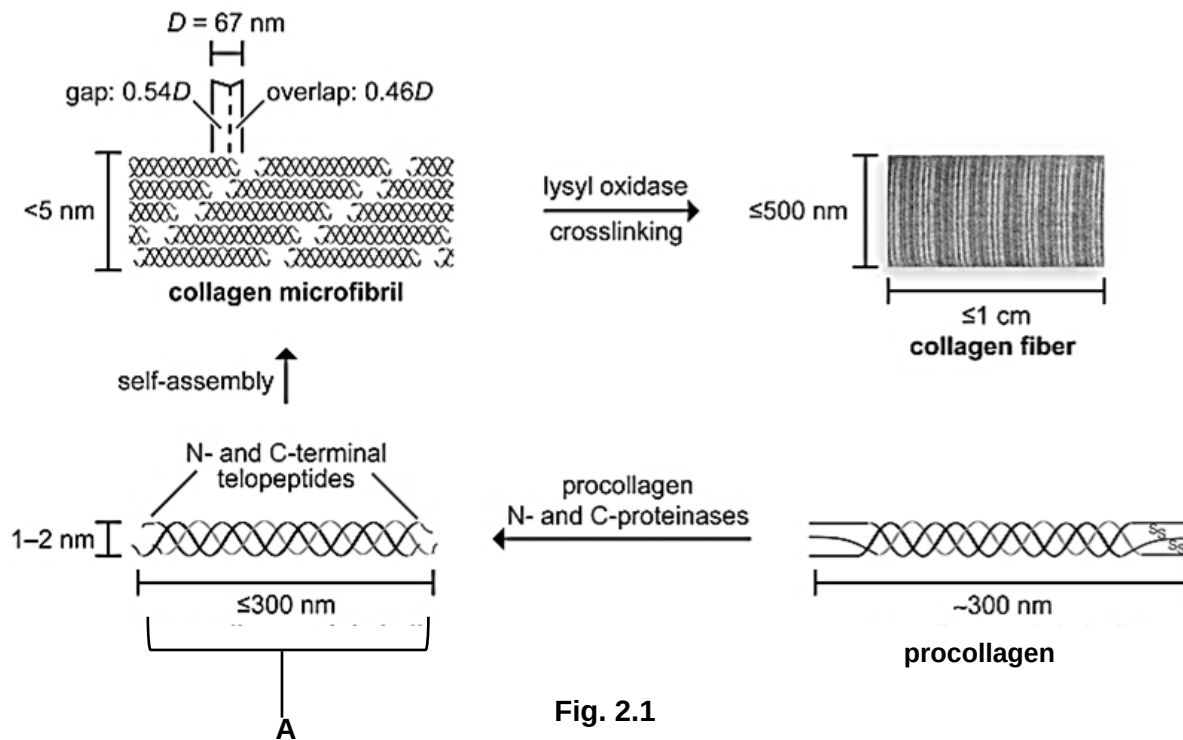


Fig. 2.1

(a) With reference to Fig. 2.1,

(i) describe structure **A**.

[2]

(ii) explain how the three chains in **A** are held together.

[1]

Modifications of the procollagen include hydroxylation, glycosylation, and disulfide bond formation. Hydroxylysines are glycosylated by the addition of galactose and elongated by the addition of glucose and galactose units.

- (iii) Distinguish **two** main structural features in the galactose-glucose–galactose chain and a single chain found in procollagen.

[2]

- (iv) Explain the need for disulfide bond formation in the modified procollagen.

[1]

- (v) State precisely the location in the cell where modifications to procollagen occur.

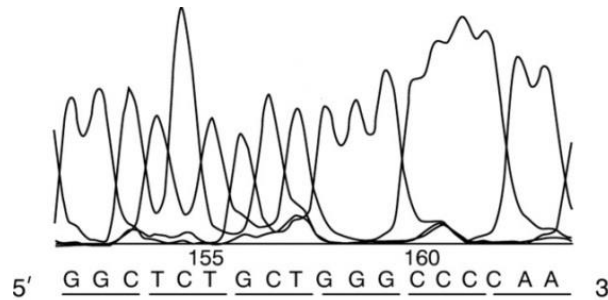
[1]

Osteogenesis imperfecta (OI) is a heritable connective tissue disorder characterised by varying degrees of bone fragility. OI is caused by mutations in one of the two α chains of type I collagen, the principal component of the extracellular matrix in bone and skin.

OI was found in a 9-year-old Turkish boy, whose parents are healthy and have no history of fractures. The boy exhibited a high fracture rate and his limb deformities have progressed to include a severely deformed pelvis, thin and markedly deformed long bones.

Sequencing of the normal and mutant DNA coding alleles was performed and the results are shown in Fig. 2.2.

Normal allele:



Mutant allele:

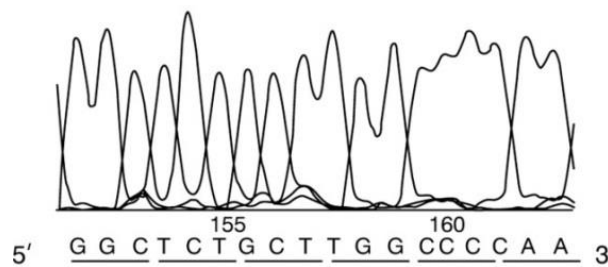


Fig. 2.2

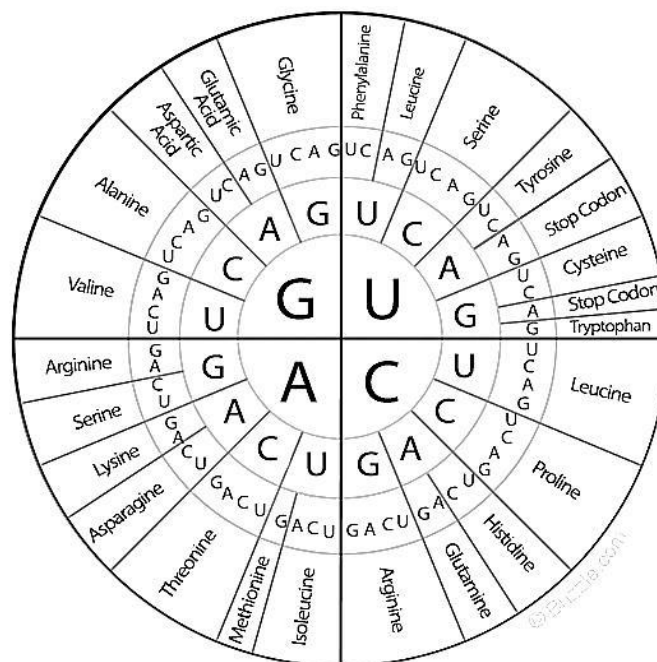


Fig. 2.3

- (b)** Using the information in Fig. 2.2 and the genetic code in Fig. 2.3, explain why the boy has a high fracture rate.

[3]

[Total: 10]

- 3 (a) Distinguish between the reproductive cycles of the influenza virus and the Human Immunodeficiency Virus (HIV).

Stage	Influenza virus	HIV
Penetration		
Replication of viral nucleic acids and integration		
Assembly and release		

[4]

SAM domain and HD domain-containing protein 1 (SAMHD1) is a protein that in humans, is encoded by the *SAMHD1* gene. SAMHD1 is a cellular enzyme, responsible for blocking replication of HIV in macrophages. It is an enzyme that exhibits phosphohydrolase activity, converting nucleotide triphosphates (NTPs) to triphosphate and a nucleoside (i.e. nucleotides without a phosphate group).

Fig. 3.1 shows the action of SAMHD1 and other cellular enzymes that target nucleic acids, on HIV replication.

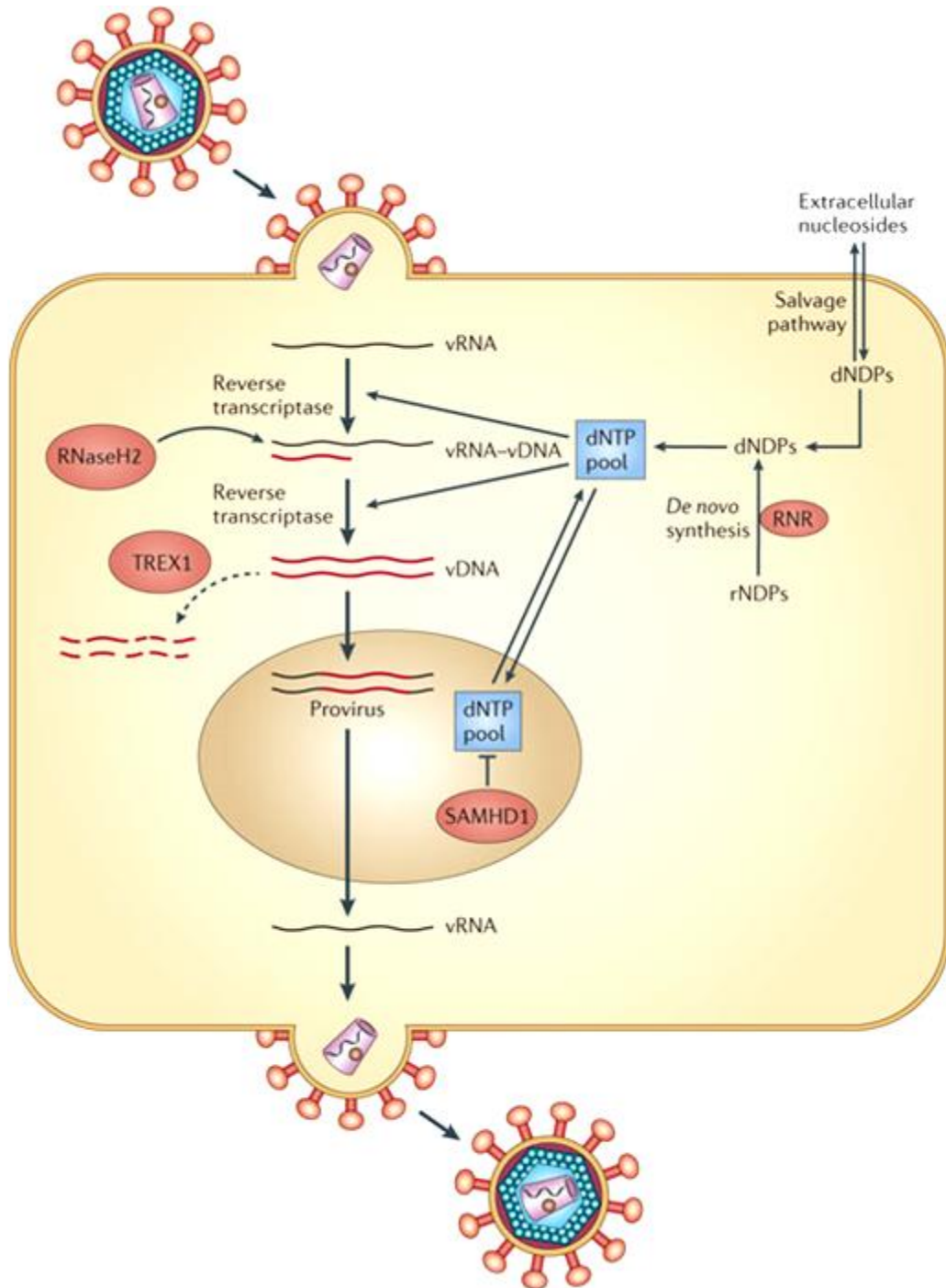


Image from *Nature Reviews Microbiology* **10**, 675-680 (2012)

Fig. 3.1

- (b) Using the information provided in the preamble and Fig. 3.1, explain how the action of SAMHD1 reduces the infectivity of HIV.

[3]

Fig. 3.2 shows the changes in HIV proteins concentration and CD4 T-cells count in the bloodstream of an individual after HIV infection.

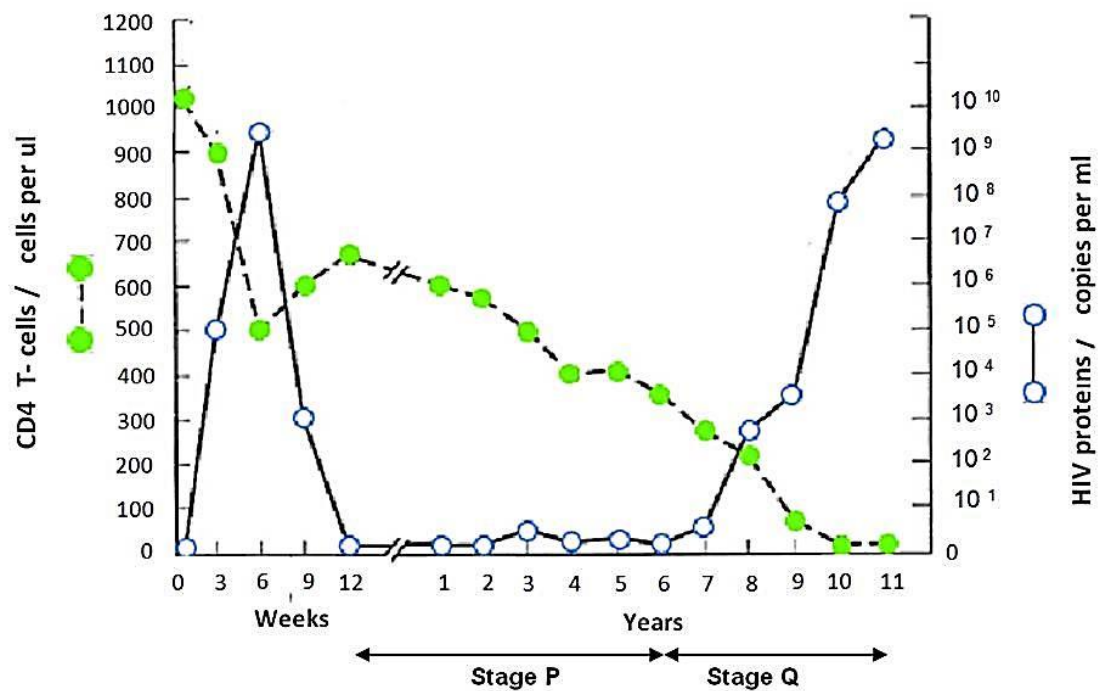


Fig. 3.2

- (c) (i) In an experiment, it was found that the concentration of the enzyme, histone deacetylase (HDAC) in infected cells at Stage **P** was higher than in infected cells not at Stage **P**.

Suggest an explanation for the presence of abnormally high levels of HDAC in infected cells at Stage **P**.

[2]

- (ii) With reference to Fig. 3.2, describe and explain the changes in the amount of HIV proteins and CD4 T-cells count during Stage **Q**.

[2]

[Total: 11]

- 4 Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 4.1 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

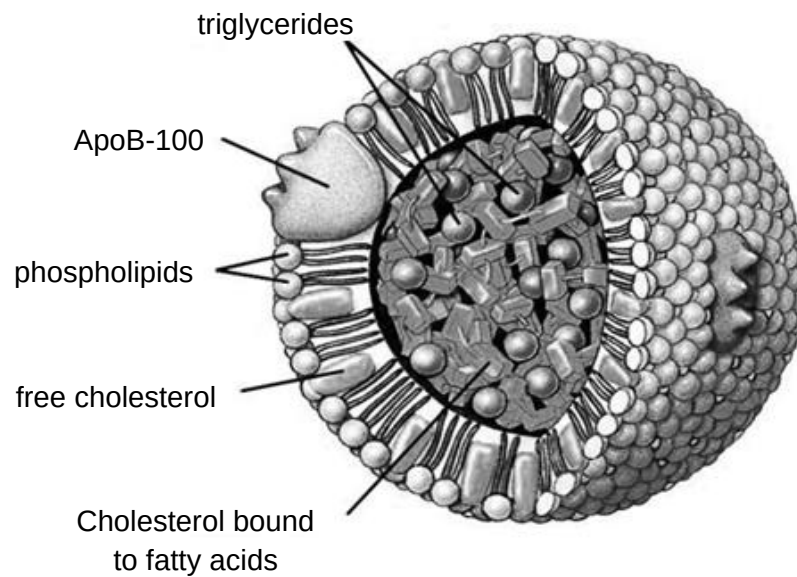


Fig. 4.1

- (a) With reference to Fig. 4.1, outline the roles of lipids in the LDL coat.

[3]

A genetic form of extremely high levels of LDL cholesterol known as familial hypercholesterolaemia, leads to severe coronary artery disease and, often, premature death. A gene and its protein product, proprotein convertase subtilisin kexin type 9, or *PCSK9*, has been revealed to play a role in regulating LDL cholesterol levels in the blood. Patients with hypercholesterolaemia had a mutation in the *PCSK9* gene which leads to increased production of *PCSK9*.

Fig. 4.2 shows the regulatory pathway which *PCSK9* participates in.

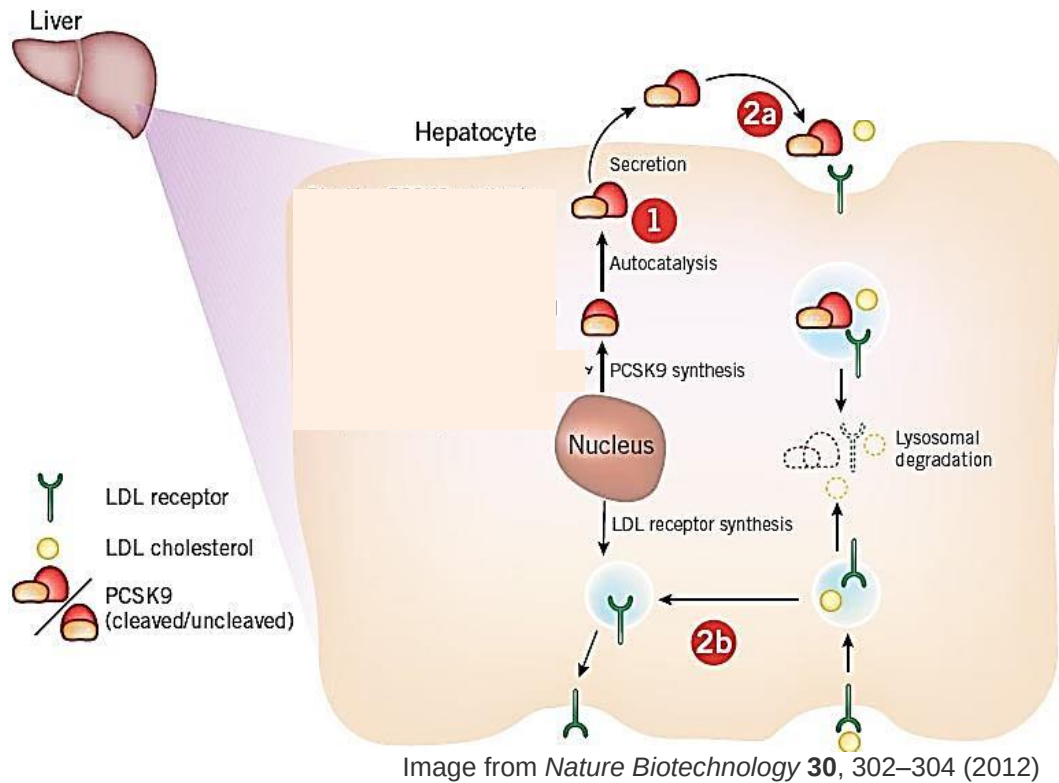


Fig. 4.2

- (b) With reference to Fig. 4.2, explain why individuals with increased *PCSK9* production will have elevated levels of LDL cholesterol in the blood.

[4]

- (c) Using the information provided in Fig. 4.2, suggest how drug therapy can be used to inhibit the action of *PCSK9*, so that cholesterol levels can be lowered in patients with familial hypercholesterolaemia.

[2]

[Total: 9]

- 5** The capsules of shepherd's purse occur in two different shapes, i.e., triangular and top-shaped. When a pure breed plant with triangular capsule is crossed with another pure breed plant having top-shaped capsule, all F_1 have only triangular character.

The F_1 offspring were selfed and the following F_2 results were obtained:

1468	triangular-shaped capsules
87	top-shaped capsules

- (a)** Using suitable symbols, draw a genetic diagram to explain the results of the F_2 generation.

[4]

(b) Explain the genetic basis of this type of inheritance.

[3]

The chi-squared (χ^2) test was performed on the results of the second cross, giving a chi-squared calculated value of 2.73.

Part of the critical values of the chi-squared distribution is shown below.

Degrees of freedom	Probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) Using the information given thus far,

(i) determine the degree of freedom.

[1]

(ii) state what conclusion may be drawn regarding the observed F_2 data.

[2]

[Total: 10]

- 6 Fig. 6.1 shows some steps involved in glycolysis and the Krebs cycle. Some ATP are made directly. Hydrogen is also released and this can result in the production of more ATP.

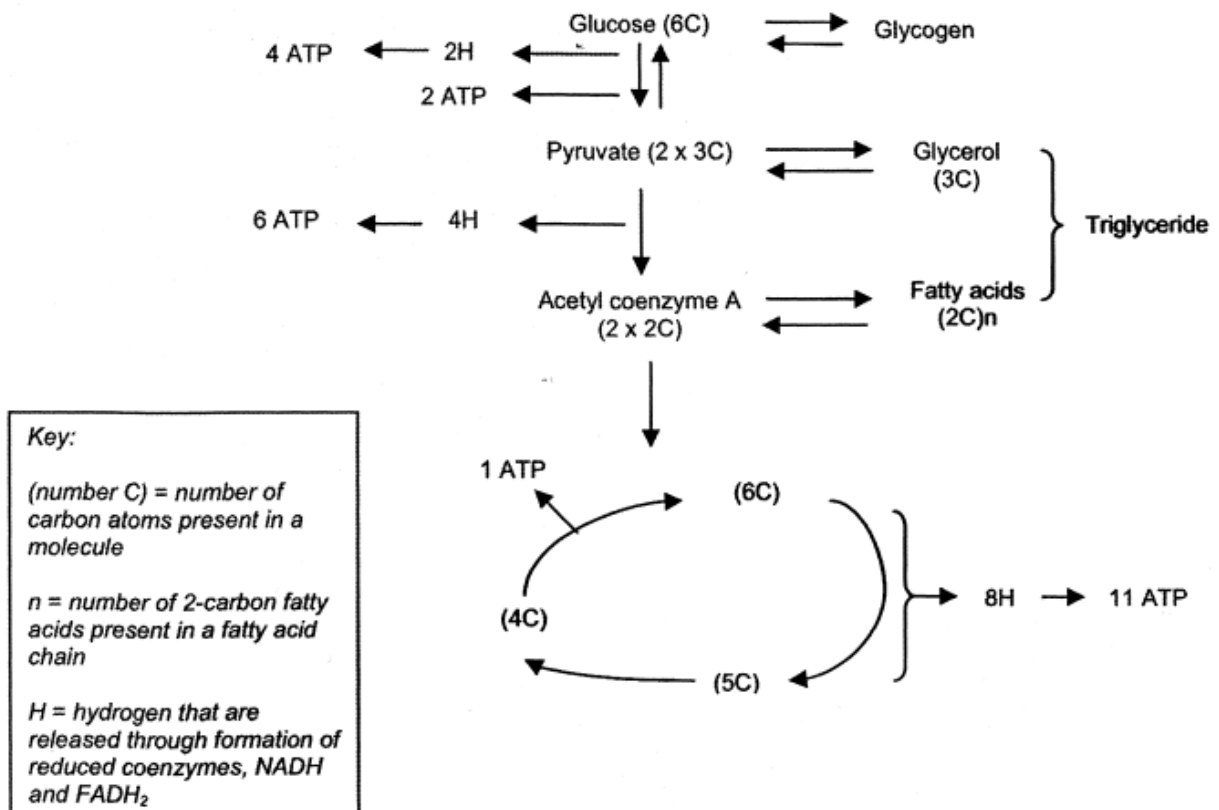


Fig. 6.1

- (a) Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP.

[4]

Similar to glycogen, triglycerides can be respired. The first step is to hydrolyse each triglyceride molecule. Each fatty acid formed is broken down into acetyl CoA molecules. The acetyl CoA molecules then enter the Krebs cycle.

- (b) Using the information from Fig. 6.1, suggest why fatty acids can only be respired under aerobic conditions.

[2]

- (c) In anaerobic respiration in muscle tissues, the conversion of glucose to lactic acid can be represented as a four-stage process shown in Fig. 6.2.

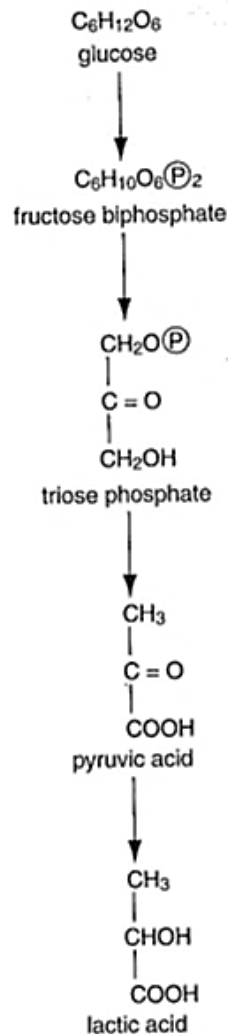


Fig. 6.2

- (i) Explain whether there is any net change in the concentration of coenzymes in the process.

[2]

- (ii) If one glucose is converted to two lactic acid molecules, state and explain for the net synthesis of ATP.

[2]

[Total: 10]

- 7 **Fig. 7.1** shows the plant cell signalling mechanism, which is comparatively less well-studied than that of the animal cell.

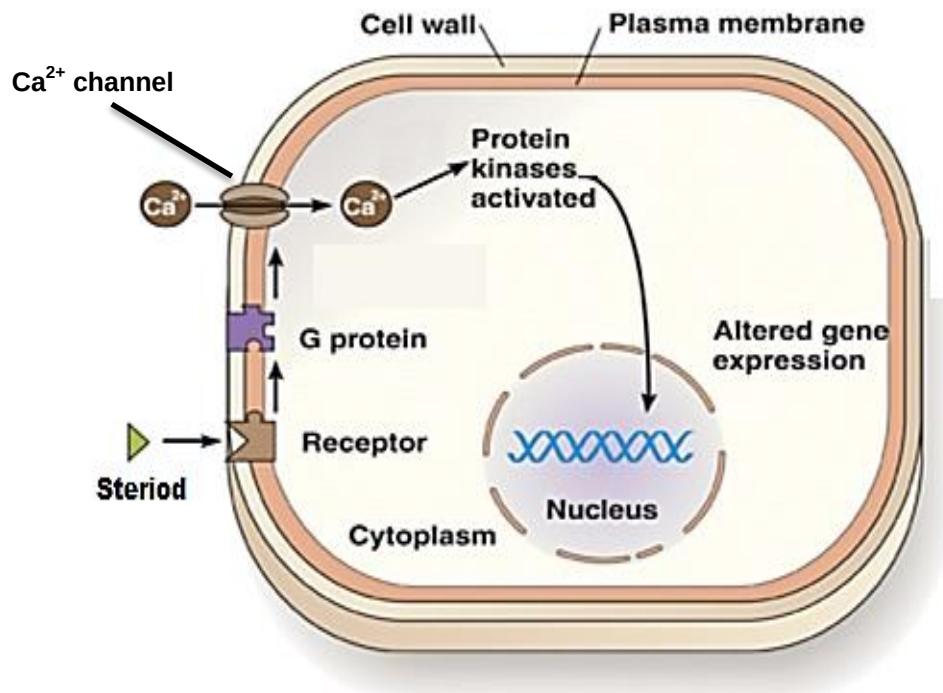


Fig. 7.1

- (a) (i) State the main stages of *cell signalling*.

[2]

- (ii) With reference to Fig. 7.1, describe the signal transduction process within the plant cell upon steroid binding.

[4]

(b) Explain how dysregulation in cell signalling contributes to cancer progression.

[3]

(c) Suggest why cancer in plants are often less consequential as compared to cancer in animals.

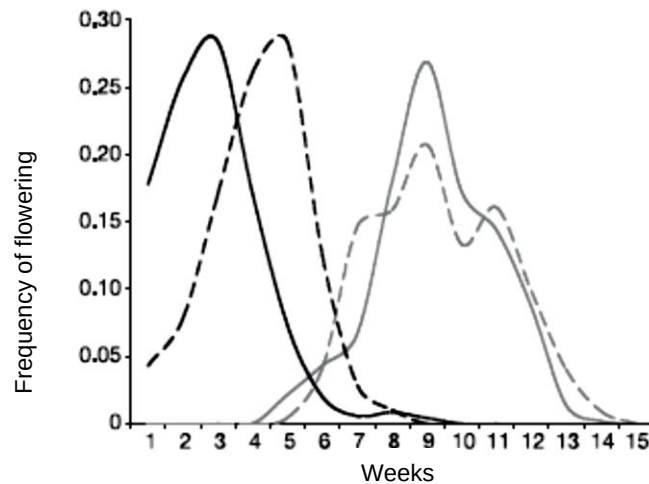
[1]

[Total: 10]

- 8 Lord Howe Island is a small landmass of less than 15 km², 580 km off the eastern coast of Australia.

Two closely related species of palm, *Howea forsteriana* and *Howea belmoreana*, are strikingly different, yet they co-exist in many parts of the island. Both are wind pollinated and diploid ($2n = 32$). Hybrids are uncommon. The two species have both female and male flowers on each individual tree. It is estimated that the two *Howea* species diverged less than one million years ago.

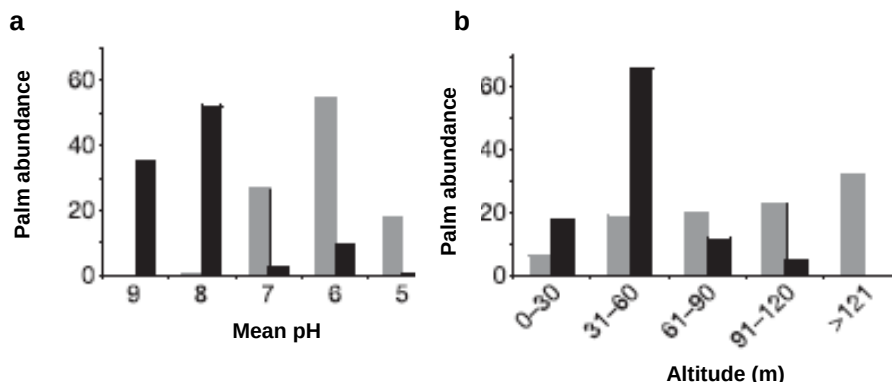
Fig. 8.1 and Fig. 8.2 show the data that have been collected concerning the frequency of flowering and the distribution of the two species respectively.



Flowering frequency for each *Howea* species. *H. belmoreana* is shown in grey, *H. forsteriana* is shown in black, with male flowering (solid line) and female flowering (dotted line) phases.

Image from *Nature* **441**, 210-213 (2006)

Fig. 8.1



Distribution of *H. forsteriana* (black) and *H. belmoreana* (grey), according to mean pH per quadrat in (a) and altitude in (b). Number of palm trees; *H. forsteriana* is shown in black ($n = 1,677$) and *H. belmoreana* in grey ($n = 4,542$). **a, b**, Distribution according to the mean pH of the soil (a) in each quadrat and elevation (b). *H. forsteriana* is found in sites at lower elevation and higher soil pH than *H. belmoreana*, and is largely restricted to calcareous soil. However, both species coexist over a large range of elevations and soil pH.

Image from *Nature* **441**, 210-213 (2006)

Fig. 8.2

- (a) With reference to the data in Fig. 8.1 and Fig. 8.2, explain how sympatric speciation may have occurred in *Howea*.

[5]

- (b) Discuss the reasons why the speciation event between *H. forsteriana* and *H. belmoreana* is more likely to be sympatric than allopatric.

[3]

- (c) *Howea belmoreana* goes by the common names, curly palm, kentia palm and Belmore sentry palm while *Howea forsteriana* goes by the names thatch palm and kentia palm.

Explain why the scientific name is better than the common names.

[2]

[Total: 10]

Section B

Answer any **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

- 9** **(a)** Compare repressible and inducible systems of gene regulation. [7]
- (b)** Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [8]
- (c)** Explain how the structures of centromeres are linked to their roles. [5]
- [Total: 20]
-
- 10** **(a)** Compare binary fission with mitosis. [7]
- (b)** Explain the control mechanisms involved in the expression of the eukaryotic genes at the translational and post-translational level. [8]
- (c)** Explain the different consequences of chromosomal mutation. [5]
- [Total: 20]

END OF PAPER